```
100 'RECTANGULAR-TO-POLAR & POLAR-TO-RECTANGULAR CONVERSION
110 '
120 DIM REX[256]                    'REX[ ]      holds the real part
130 DIM IMX[256]                    'IMX[ ]      holds the imaginary part
140 DIM MAG[256]                    'MAG[ ]      holds the magnitude
150 DIM PHASE[256]                  'PHASE[ ] holds the phase
160 '
170 PI = 3.14159265
180 '
190 GOSUB XXXX                      'Mythical subroutine to load data into REX[ ] and IMX[ ]
200 '
210 '
220 '                               'Rectangular-to-polar conversion, Eq. 8-6
230 FOR K% = 0 TO 256
240   MAG[K%] = SQR( REX[K%]^2 + IMX[K%]^2 )          'from Eq. 8-6
250   IF REX[K%] = 0 THEN REX[K%] = 1E-20             'prevent divide by 0 (nuisance 2)
260   PHASE[K%] = ATN( IMX[K%] / REX[K%] )            'from Eq. 8-6
270   '                                               'correct the arctan (nuisance 3)
280   IF REX[K%] < 0 AND IMX[K%] <   0 THEN PHASE[K%] = PHASE[K%]  - PI
290   IF REX[K%] < 0 AND IMX[K%] >= 0 THEN PHASE[K%] = PHASE[K%] + PI
300 NEXT K%
310 '
320 '
330 '                               'Polar-to-rectangular conversion, Eq. 8-7
340 FOR K% = 0 TO 256
350   REX[K%] = MAG[K%] * COS( PHASE[K%] )
360   IMX[K%] = MAG[K%] *  SIN( PHASE[K%] )
370 NEXT K%
380 '
390 END
```

TABLE 8-3

Nuisance 2: Divide by zero error

When converting from rectangular to polar notation, it is very common to find frequencies where the real part is zero and the imaginary part is some nonzero value. This simply means that the phase is exactly 90 or -90 degrees. Try to tell your computer this! When your program tries to calculate the phase from: $Phase\,X[k] = \arctan(Im\,X[k] / Re\,X[k])$, a *divide by zero error* occurs. Even if the program execution doesn't halt, the phase you obtain for this frequency won't be correct. To avoid this problem, the real part must be tested for being zero before the division. If it is zero, the imaginary part must be tested for being positive or negative, to determine whether to set the phase to $\pi/2$ or $-\pi/2$, respectively. Lastly, the division needs to be bypassed. Nothing difficult in all these steps, just the potential for aggravation. An alternative way to handle this problem is shown in line 250 of Table 8-3. If the real part is zero, change it to a negligibly small number to keep the math processor happy during the division.

Nuisance 3: Incorrect arctan

Consider a frequency domain sample where $Re\,X[k] = 1$ and $Im\,X[k] = 1$. Equation 8-6 provides the corresponding polar values of $Mag\,X[k] = 1.414$ and $Phase\,X[k] = 45°$. Now consider another sample where $Re\,X[k] = -1$ and